# D1.3 Mutation and gene editing (HL)

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 30 |

**About 30 minutes.** The number of marks shows how many separate points to make.

## Section A: Recall

6 MARKS

1 **DEFINE** gene knockout. [D1.3.8] [1]

2 **STATE** two features that make the mouse a useful model organism. [D1.3.8] [2]

3 **STATE** the role of Cas9. [D1.3.9] [1]

4 **DISTINGUISH** between conserved and highly conserved sequences. [D1.3.10] [2]

## Section B: Understanding

12 MARKS

5 **EXPLAIN** how a knockout mouse can show the function of a gene. [D1.3.8] [3]

6 **OUTLINE** how CRISPR-Cas9 is used to edit a chosen gene. [D1.3.9] [3]

7 **DESCRIBE** one successful use of CRISPR-Cas9. [D1.3.9] [2]

**8** **EXPLAIN** two hypotheses for why some sequences are highly conserved. [D1.3.10] [4]

## Section C: Working with evidence

8 MARKS

*All data in this section are invented for practice. The patterns are realistic.*

**9** [D1.3.8]

Mice with one gene knocked out (KO) were compared with normal (wild-type) mice of the same age, fed the same diet.

| MEASUREMENT AT 16 WEEKS | WILD-TYPE (N = 12) | KNOCKOUT (N = 12) |
|---|---|---|
| BODY MASS / G | 28 ± 2 | 52 ± 4 |
| FOOD EATEN PER DAY / G | 3.9 ± 0.3 | 6.8 ± 0.5 |
| BLOOD GLUCOSE / MMOL $DM^{-3}$ | 7.1 ± 0.6 | 12.4 ± 1.5 |

**a** **CALCULATE** the percentage difference in body mass of the knockout mice compared with wild-type. [2]

**b** **SUGGEST** a function of the knocked-out gene, using the data. [2]

**c** **EXPLAIN** why the diet and age of the mice were kept the same. [2]

**d** **DISCUSS** whether the results show that the same gene controls appetite in humans. [2]

## Section D: Longer answer

4 MARKS

**10** **EVALUATE** the use of gene knockout in model organisms to study human genes. [D1.3.8] [4]